Shorter Races and Carb in the Mouth

Endurance Fueling Guide

As a coach and physiotherapist, I have always believed in empowering athletes with knowledge. Not just so they understand the advice and actually follow it, but more importantly so they truly believe in what they are doing.

When athletes understand the science and rationale behind their fueling strategy, they show up on training days and race days with more confidence, and that confidence help them perform better.

Fueling is not just about avoiding bonking. Done well, it helps you hold pace, protect training quality, recover faster, and reduce the risk of race-day mistakes.

The goal is not to copy exactly what elite athletes do. The goal is to build a fueling strategy based on real-world practice and modern sports nutrition science, while matching your event or training session, your gut, your body, and your conditions.

This Science Hub, together with your automatically created fueling schedules and plans, gives you the knowledge you need to take ownership of your fueling in a simple way and perform at your best.

/Göran

How Much Carbohydrate Do You Need?

Carbohydrate needs depend mostly on duration, intensity, and gut tolerance.

Recommended Carbohydrate Intake by Duration

SESSION / RACE	PRACTICAL CARB TARGET
Under 60 min	Usually little or none needed
60–120 min	~30–60 g/hour
2–3 hours	~60–90 g/hour
3+ hours	~90 g/hour or more, if trained
Elite / highly gut-trained athletes	**100–120+ g/hour may be possible**

Classic sports nutrition guidelines recommend 30–60 g/h for endurance exercise lasting 1–2.5 hours, and up to 90 g/h for longer events when using multiple transportable carbohydrates such as glucose + fructose. JEUKENDRUP 2014

More recently, elite endurance sport has pushed beyond this. In cycling, marathon running, trail running, and triathlon, many athletes now use 90–120 g/h, and sometimes more. But higher is not automatically better. The best target is the highest useful amount you can tolerate consistently. CAO ET AL. 2025 REVIEW

> **PRACTICAL TAKEAWAY**
> Start with what you tolerate now, then build gradually. For many runners, moving from 30g/h to 60–90g/h is already a huge upgrade.

Shorter Races and Carb in the Mouth

For shorter races like a 5K or 10K, you usually do not need large amounts of carbohydrate during the race. If you start well fueled, your stored glycogen is normally enough for the distance.

But small carbohydrate exposure in the mouth may still help in some situations.

Water Only

Baseline effort & motor output

Carb Rinse / Strip

Receptors signal brain → potential perceived effort drop

This is called a **carbohydrate mouth rinse**. The idea is not mainly that the carbs fuel the muscles directly. Instead, carbohydrate receptors in the mouth may signal to the brain that fuel is available, potentially influencing perceived effort, motivation, and motor output.

The evidence is mixed. A 5K running study found no clear dose-response benefit from different carbohydrate mouth-rinse concentrations in recreational runners. CLARKE ET AL. 2017 But newer research found that carbohydrate dissolvable strips improved performance in a 12.87 km treadmill time trial compared with water, while carbohydrate mouth rinse improved some pacing and performance markers without increasing perceived effort. GUADAGNI ET AL. 2026 A 2025 systematic review and meta-analysis also suggests carbohydrate mouth rinsing can be ergogenic, although the effect is likely small and context-dependent. DENG ET AL. 2025 META-ANALYSIS

For training, this is mostly relevant for hard shorter sessions where you want high quality without taking in a lot of fuel, such as 5K/10K workouts, VO2max intervals, or race-pace sessions. A small carb sip or rinse before/during the session may help some athletes feel sharper, especially if training early, slightly underfueled, or mentally flat. Actually fueling some of these harder sessions with carbs can also make you feel better during the workout, give you another chance to train your gut, and may reduce how much you deplete glycogen, which can help recovery for the next session. You don't need huge amounts for a short workout, but for key sessions, a little fuel can be a smart performance and recovery tool.

PRACTICAL TAKEAWAY

For a 5K, 10K, or short hard workout, you usually do not need meaningful carb intake during the effort. But a small carb sip, mouth rinse, or dissolvable carb strip can be worth testing as a marginal-gain tool, especially for hard efforts lasting around 20–60 minutes. It is not the foundation of fueling, but it may help some athletes get a little more out of short races and key sessions.

The High-Carb Revolution

For years, 60g/h was seen as a high carbohydrate intake. Now many elite athletes are fueling with 90–120g/h or more.

In the Breaking2 project, Eliud Kipchoge reportedly used around 100g/h. In more recent elite marathon examples, Sabastian Sawe's sub-2-hour attempt fueling was reported at around 115g/h. In pro cycling, fueling around 100–120g/h has become much more common in hard races and stages. JEUKENDRUP 2014

One of the most interesting studies compared 60, 90, and 120g/h during a mountain marathon in elite trail runners. The 120g/h group showed lower markers of muscle damage and better recovery of high-intensity running capacity 24 hours later. This does not mean every runner should take 120g/h, but it shows what may be possible when the gut is trained. VIRIBAY ET AL. 2020

Muscle Damage Markers Post-Race (Relative Concept)

A later study also found that 120g/h preserved neuromuscular function and improved next-day high-intensity capacity after a trail marathon. URDAMPILLETA ET AL. 2020

PRACTICAL TAKEAWAY

High-carb fueling is not just race-day energy. For long or hard sessions, it may also help

Gut Training

Your gut is trainable.

If you suddenly try to take 90–120g/h on race day without practice, the risk of stomach problems is high. But if you regularly practice taking carbs during training, many athletes improve tolerance and reduce GI symptoms over time. JEUKENDRUP 2017

A simple gut-training progression:

1. **Start Baseline**
   Start with your current comfortable intake
2. **Gradual Increase**
   Add 5–15g/h gradually over several key sessions
3. **Race Specificity**
   Practice during long runs and race-specific workouts
4. **Final Testing**
   Test your exact race products or DIY mix before race day
5. **Race Day Rule**
   Do not try a new high-carb strategy for the first time in a race

Gut training is especially important for runners because running creates more gut movement and impact than cycling, which can make GI tolerance more challenging.

PRACTICAL TAKEAWAY

Your race gut is built in training. Use long runs as your fueling lab.

Health Concerns

It is reasonable to ask whether taking in large amounts of sugar during training and racing is healthy.

The key context is that sugar during hard or long exercise is not the same as sugar while sitting still. During exercise, contracting muscles rapidly increase glucose uptake, partly through insulin-independent GLUT4 translocation. In simple terms: your working muscles are actively pulling carbohydrate from the blood and using it as fuel. Reviews suggest skeletal muscle is responsible for the majority of whole-body glucose disposal, and exercise can increase muscle glucose uptake dramatically compared with rest. EVANS ET AL. 2019

This is one reason regular endurance exercise is associated with better insulin sensitivity and lower metabolic disease risk. So for a healthy endurance athlete, carbohydrate taken during long or hard exercise is usually being used to support performance, maintain blood glucose, reduce excessive glycogen depletion, and improve training quality. JEUKENDRUP 2014

That does not mean everyone should drink sugar all day. High-carb fueling makes most sense during sessions where performance, duration, or recovery matter: long runs, races, hard workouts, and big training days. For short easy runs, many athletes need little or no fuel during the session.

PRACTICAL TAKEAWAY

High-carb fueling is not about adding random sugar to your diet. It is about placing carbohydrate where your body can use it best: during the work. Fuel key sessions well, then build the rest of your diet around normal, nutrient-rich foods.

Glucose, Fructose, and Transporters

The reason modern fueling often combines different carbohydrates is simple: your gut has different transport pathways.

Glucose and maltodextrin mainly use the **SGLT1 transporter**. Fructose mainly uses the **GLUT5 transporter**. Combining glucose/maltodextrin with fructose lets you use both pathways and can increase the amount of carbohydrate you absorb and oxidize during exercise. JEUKENDRUP 2010

Max Absorption Rates (g/hour) vs Transport Pathways

Older research showed that glucose-only fueling tends to plateau around 1.0–1.1 g/min of exogenous carbohydrate oxidation, while glucose + fructose can push oxidation higher. A classic study also found better cycling time-trial performance with glucose + fructose compared with glucose alone. PUBMED

2:1 Ratio

The classic endurance fueling ratio. Often works well and may suit athletes who are more sensitive to fructose.

1:0.8 Ratio

Common in many modern high-carb fueling products, especially when aiming for higher carb intakes.

1:1 Ratio

This is what you get from table sugar (sucrose). It is cheap, simple, and works very well for many athletes.

PRACTICAL TAKEAWAY

The exact ratio matters most when carb intake is high. For many runners, total carbs, gut training, drink concentration, and timing matter just as much.

Concentration & Timing

Carbohydrate amount is only one part of the plan. Concentration and timing matter too.

Very concentrated drinks or gels can sit heavily in the stomach or pull water into the gut, increasing the risk of bloating, nausea, or diarrhea. Maltodextrin can help because it provides glucose units with lower osmolality than the same amount of simple sugar, which may improve tolerance in some high-carb mixes. PÉREZ-CASTILLO ET AL. 2023

Small, frequent doses usually work better than large gaps followed by panic fueling. For many runners, taking carbs every 15–25 minutes is a practical rhythm.

PRACTICAL TAKEAWAY

If your stomach struggles, do not only reduce carbs. Also check concentration, timing, fluid intake, fructose amount, and race intensity.

Sodium and Electrolytes

The main electrolyte lost in sweat is sodium.

Typical Sweat Electrolyte Composition (mg/L)

Potassium, magnesium, and calcium matter for general health, but during endurance exercise sodium is usually the main electrolyte relevant for fluid balance. Sodium losses vary a lot depending on sweat rate, heat, body size, fluid intake, and individual sweat sodium concentration. MCDERMOTT ET AL. 2017 / NATA

Sodium can help with fluid retention and taste, but it is not magic. Taking sodium does not fully protect against exercise-associated hyponatremia if you drink too much fluid. Overdrinking is the main risk factor. HEW-BUTLER ET AL. 2017

For most runners, the goal is not to replace 100% of sodium lost during the event. The goal is to take in enough fluid and sodium to support performance without overhydrating. SAWKA ET AL. 2007 / ACSM

What About Potassium or Magnesium?

Potassium is also lost in sweat, but usually in much smaller amounts than sodium. Many sports drinks include a small amount of potassium, and ACSM describes typical fluid-replacement drinks as containing about 2–5 mmol/L potassium, or roughly 80–200 mg/L. SAWKA ET AL. 2007 / ACSM

For races over 3–4 hours, adding a small potassium source can be reasonable, but it is optional, not essential. A simple DIY option is 100 ml apple juice per 500 ml bottle. This adds flavor, some carbs, and potassium. The app has this apple juice option integrated in the DIY sports drink and gel builders.

Magnesium and calcium also matter for general health, but they are not core during-exercise fueling targets. Sweat losses are much smaller than sodium, and we do not recommend adding them to the race drink by default. MONTAIN ET AL. 2007

PRACTICAL TAKEAWAY

Sodium matters most in long, hot, sweaty sessions. But more is not always better.

Hydration

Hydration needs are highly individual.

Fixed advice like "drink X ml every hour" is often too generic because sweat rate can vary massively between athletes and conditions. Larger athletes, hotter weather, higher intensity, and longer duration usually increase fluid needs. MCDERMOTT ET AL. 2017 / NATA

A good hydration plan should help you avoid both extremes:

- **Too little fluid**, which can impair performance and increase heat strain
- **Too much fluid**, which can increase the risk of hyponatremia

A practical approach is to start hydrated, drink regularly during longer sessions, adjust to heat and sweat rate, and avoid gaining body weight during endurance events.

PRACTICAL TAKEAWAY

Hydration should be planned, but not forced. Drink enough to perform, not so much that you override your body.

Carb Loading

Carb loading is most useful before longer races where glycogen depletion can limit performance, such as marathons, long trail races, ultras, triathlons, and long cycling events.

A common evidence-based target is around **8–12g carbohydrate per kg body weight per day** for the final 24–48 hours before a long endurance event, combined with reduced training. WALLIS 2025 / GSSI

A pre-race meal of around 1–4g/kg carbohydrate 1–4 hours before competition is also commonly recommended, depending on timing and gut tolerance. WALLIS 2025 / GSSI

Pre-Race Carb Loading Targets	
Days 1-2 pre-race	8-12 g/kg/day
1-4 hours pre-race meal	1-4 g/kg

Keep it simple:

- choose familiar high-carb foods
- reduce fiber slightly if your gut is sensitive
- avoid huge last-minute meals
- spread carbs across the day
- do not rely only on one massive pasta dinner

PRACTICAL TAKEAWAY

Carb loading fills the tank. Fueling during the race keeps the engine supplied.

DIY Fueling

Many sports drinks and gels are built from simple ingredients:

- carbohydrate
- sodium
- water
- flavoring

That means homemade fueling can work extremely well if the amounts are correct.

A simple sugar + salt mix can be effective because table sugar provides both glucose and fructose. More advanced DIY mixes can use maltodextrin and fructose to match common commercial ratios like 1:0.8 or 2:1. The scientific logic is the same: combine carbohydrate types to improve absorption and oxidation at higher intakes. JEUKENDRUP 2010

Estimated Cost per Hour (90g carbs)

The main things to get right are:

- carbs per hour
- glucose/fructose ratio
- fluid amount
- sodium amount
- concentration
- timing

One downside of frequent sugar-based fueling is dental exposure, especially if the drink or gel is acidic and sipped for a long time. Reviews on athlete oral health highlight sports drinks, sugar exposure, acidity, and reduced saliva during exercise as possible contributors to dental problems. SCHULZE ET AL. 2024

PRACTICAL TAKEAWAY

Homemade fueling can be cheap and effective, but precision still matters.

Teeth Health

The biggest real health downside of high-carb fueling is probably dental risk.

Sports drinks, gels, chews, and homemade sugar mixes can expose the teeth to frequent carbohydrate. Many commercial products are also acidic, which may increase enamel erosion risk. During exercise, the risk can be amplified because breathing hard, dehydration, and reduced saliva flow can make the mouth drier, lowering the mouth's natural protection. Reviews on athlete oral health highlight sugar exposure, acidity, sports drinks, and dry mouth during exercise as potential contributors to dental caries and erosion. SCHULZE ET AL. 2024

This does not mean you should avoid fueling important sessions. It means you should manage the exposure.

Simple ways to reduce risk:

- Use high-carb fueling mainly for sessions where it matters
- Avoid sipping sugary or acidic drinks for hours outside training
- Rinse your mouth with water after gels or sports drinks when practical
- Choose less acidic flavoring when possible
- Do not brush immediately after acidic sports drinks; rinse first and wait
- Use fluoride toothpaste and keep normal dental checkups
- For training, consider simpler DIY mixes with less acidity

A recent systematic review found mixed evidence linking sports drinks alone to dental erosion, but the overall risk depends on frequency, acidity, saliva, oral hygiene, and total exposure. GÁLVEZ-BRAVO ET AL. 2025

PRACTICAL TAKEAWAY

Fuel the sessions that matter, but protect your teeth by limiting unnecessary sugar/acid exposure outside training and rinsing with water when possible.

Legal Performance Enhancers

Fueling is the foundation. But some legal supplements can provide small but meaningful performance benefits in the right situation.

Caffeine

Caffeine is one of the best-supported legal performance aids in endurance sport.

A common effective range is 3–6mg/kg body weight, although some athletes respond well to less. The ISSN position stand states that caffeine consistently improves endurance performance when used appropriately in this range. GUEST ET AL. 2021 / ISSN CAFFEINE

More is not always better. High doses can increase anxiety, heart rate, sleep problems, and GI discomfort.

Best used for: races, key workouts, late-race focus, and longer events where fatigue builds.

Sodium Bicarbonate

Sodium bicarbonate can improve performance when high-intensity effort and acidosis are limiting factors. This may include hard intervals, repeated surges, middle-distance races, intense hill racing, or hard race finishes.

The ISSN position stand reports performance benefits with doses around 0.2–0.5g/kg body weight, but GI distress is the major practical challenge. GRGIC ET AL. 2021 / ISSN BICARBONATE

Best used for: high-intensity efforts, repeated hard surges, short-to-medium races, or races with brutal climbs/finishes.

Nitrate / Beetroot

Dietary nitrate, often from beetroot juice or nitrate salts, can improve nitric oxide availability and may reduce the oxygen cost of exercise in some athletes.

A 2025 umbrella review of 20 systematic reviews found evidence that nitrate supplementation can improve several exercise-performance outcomes, although responses vary depending on event type, training status, dose, and individual response. POON ET AL. 2025

Best used for: some endurance events, repeated high-intensity efforts, and athletes who respond well in testing.

> #### Important Note
>
> Legal performance enhancers should be tested in training before racing. The IOC consensus statement emphasizes that supplements should be used only after considering the athlete's goals, evidence, safety, and risk of contamination. MAUGHAN ET AL. 2018 / IOC

PRACTICAL TAKEAWAY

Caffeine, bicarbonate, and nitrate can help, but they are additions to good fueling, not replacements for it.

Common Fueling Mistakes

1. Underfueling long races

Many runners take far less carbohydrate than modern evidence supports, leading to late-race fading and bonking.

2. Trying new fueling on race day

Your gut needs practice. Race day should be execution, not experimentation.

3. Over-drinking

Drinking too much because "hydration is always good" can be dangerous and cause hyponatremia. HEW-BUTLER ET AL. 2015

4. Thinking it must be expensive

DIY fueling can work extremely well if calculated properly. Commercial products add convenience, not magic.

5. Ignoring normal daily nutrition

Fueling during exercise helps, but it cannot fully compensate for poor daily energy intake, low carbohydrate availability, poor sleep, or bad training.

The Big Picture

Good fueling is not about always taking the maximum amount possible.

It is about taking the right amount for:

- your event
- your intensity
- your gut
- your environment

- your goals

For short easy runs, you may need little or nothing. For long races and key sessions, proper fueling can be one of the biggest performance upgrades available.

THE BEST STRATEGY IS SIMPLE:

TRAIN THE BODY.

TRAIN THE GUT.

PRACTICE THE PLAN.

THEN RACE WITH CONFIDENCE.

Learn More: Key Studies

Carbohydrate Guidelines

Thomas et al. 2016 / AND-DC-ACSM
A major consensus paper covering carbohydrate needs, hydration, supplements, and race nutrition.

Burke et al. 2011
A key review on carbohydrate periodization, exercise fueling targets, and how carb needs vary by session duration and intensity.

Multiple Transportable Carbohydrates

Jeukendrup, Multiple transportable carbohydrates (2010)
A central review explaining why combining glucose and fructose allows higher carbohydrate oxidation than glucose alone.

Currell & Jeukendrup (2008)

A classic study showing superior endurance performance with glucose + fructose compared with glucose alone during prolonged cycling.

Cao et al., 2025 review

A modern review of carbohydrate supplementation approaches in endurance athletes, including high-carb strategies and gut tolerance.

120g/h & High-Carb Revolution

Viribay et al. (2020)

Compared 60, 90, and 120g/h in elite trail runners. The 120g/h group showed lower exercise-induced muscle damage markers.

Urdampilleta et al. (2020)

Found that 120g/h may limit neuromuscular fatigue and improve next-day high-intensity run capacity compared with 60 and 90g/h.

Gut Training and GI Symptoms

Jeukendrup, Training the Gut for Athletes (2017)

Important review explaining how the gastrointestinal system can adapt to carbohydrate intake during exercise.

Mlinaric et al., 2025 review on GI symptoms

Modern review on nutritional strategies to reduce GI symptoms in endurance athletes, including gut training and carbohydrate tolerance.

Hydration, Sodium, and Hyponatremia

McDermott et al. 2017 / NATA

Practical evidence-based guidance on hydration, dehydration, overhydration, and individual fluid planning.

Hew-Butler et al. 2015

Important consensus statement emphasizing that overdrinking is the primary cause of exercise-associated hyponatremia.

Legal Performance Enhancers

Maughan et al. 2018 / IOC

Major consensus statement on supplements, evidence, safety, and risk management for athletes.

Guest et al. 2021 / ISSN Caffeine

Detailed review of caffeine dose, timing, performance effects, and safety.

Grgic et al. 2021 / ISSN Bicarbonate

Detailed position stand on sodium bicarbonate dosing, performance effects, and GI issues.

Poon et al. 2025 / Nitrate Umbrella Review

Large umbrella review summarizing the evidence for nitrate supplementation across exercise performance outcomes.